

Directional Simulation

```
library(spatstat.geom)
```

```
## Loading required package: spatstat.data
```

```
## Loading required package: spatstat.univar
```

```
## spatstat.univar 3.1-6
```

```
## spatstat.geom 3.7-0
```

```
library(spatstat.random)
```

```
## spatstat.random 3.4-4
```

```
W <- owin(xrange = c(0, 10), yrange = c(0, 10))
```

```
sigma_parallel <- 0.60
```

```
sigma_perp <- 0.12
```

```
theta0 <- pi / 4
```

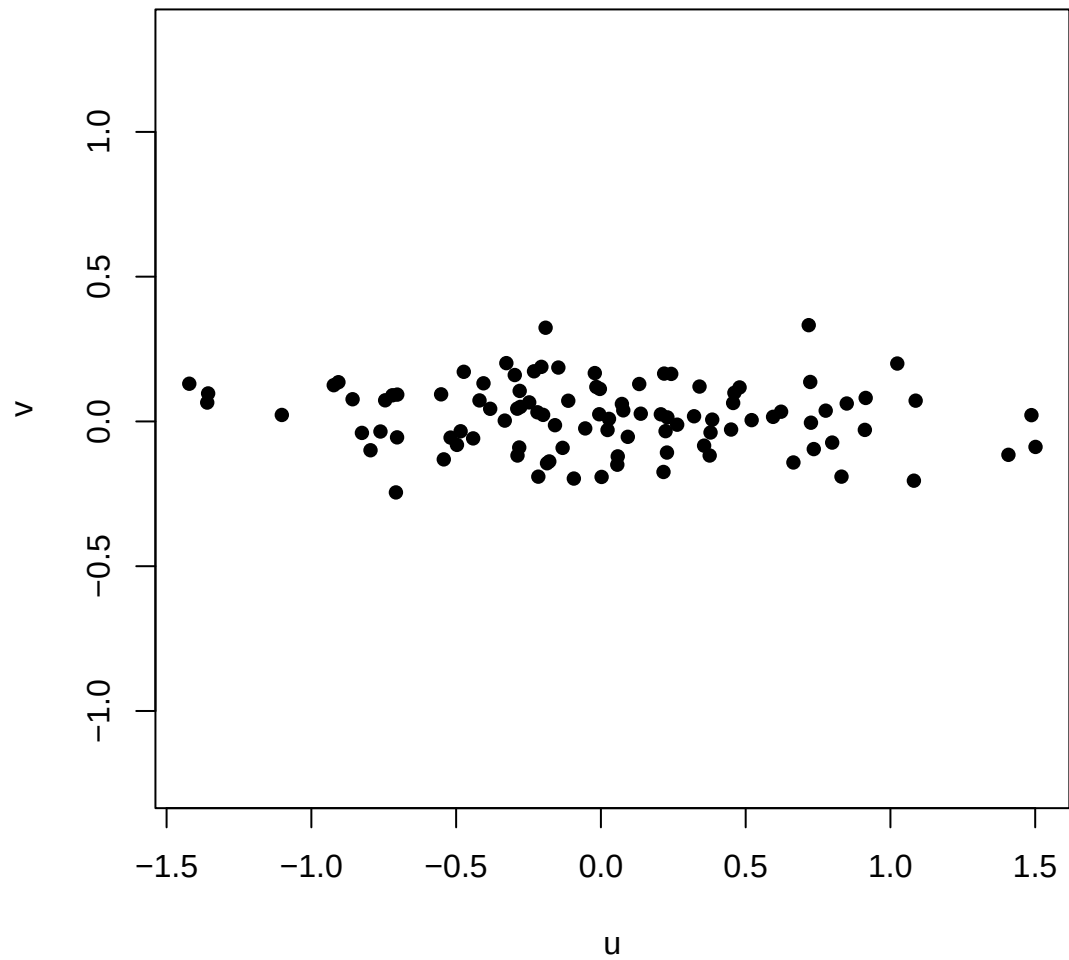
```
n <- 100
```

```
u <- rnorm(n, mean = 0, sd = sigma_parallel)
```

```
v <- rnorm(n, mean = 0, sd = sigma_perp)
```

```
plot(u, v, asp = 1, pch = 16,  
      xlab = "u", ylab = "v", main = "Before rotation")
```

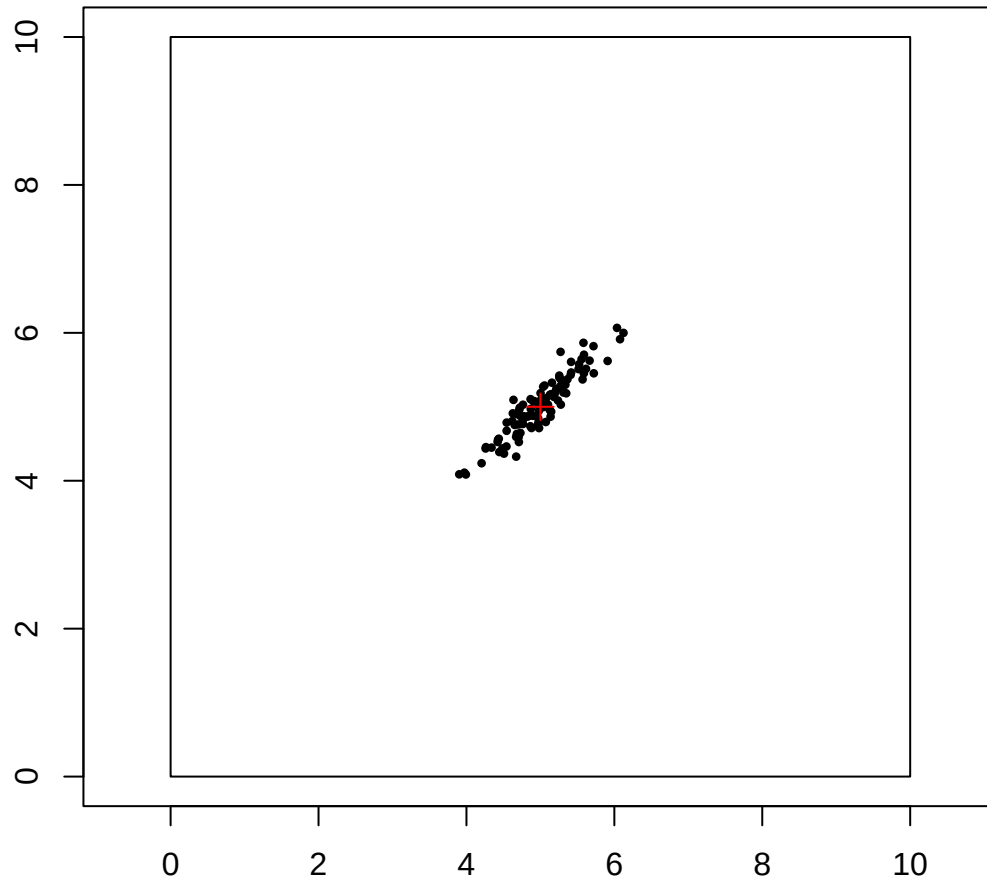
Before rotation



```
dx <- u * cos(theta0) - v * sin(theta0)
dy <- u * sin(theta0) + v * cos(theta0)
xy <- data.frame(x = 5 + dx, y = 5 + dy)

plot(W, main = "One ellipse cluster", axes = TRUE)
points(xy$x, xy$y, pch = 16, cex = 0.6)
points(5, 5, pch = 3, col = "red", cex = 1.3)
```

One ellipse cluster



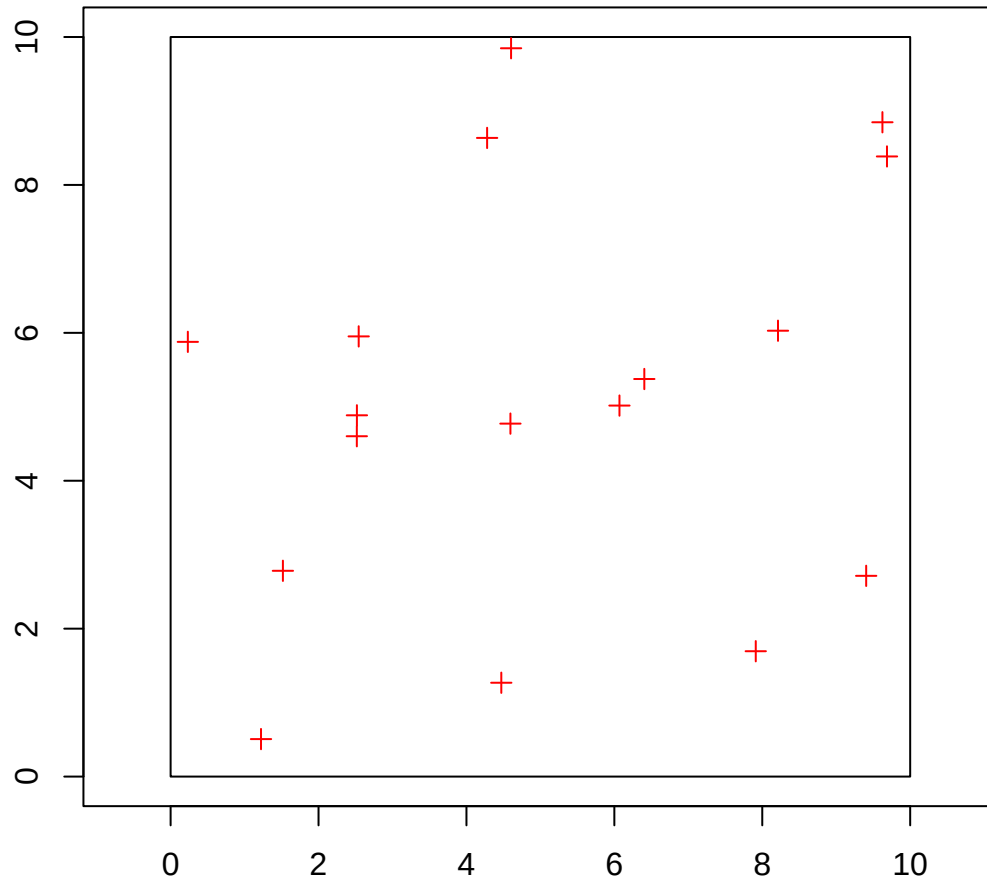
```
lambda_centers <- 0.2
X_parent <- rpoispp(lambda_centers, win = W)
xy_parent <- data.frame(x = X_parent$x, y = X_parent$y)
n_parent <- npoints(X_parent)

n_parent
```

```
## [1] 17
```

```
plot(X_parent, main = "Cluster centres", axes = TRUE,
     pch = 3, cols = "red")
```

Cluster centres



```
lambda_offspring <- 40
n_offspring <- rpois(n_parent, lambda = lambda_offspring)

n_offspring

## [1] 37 44 46 46 42 44 38 47 35 49 47 47 48 40 32 42 32

xy <- data.frame(x = numeric(), y = numeric()) #
for (i in seq_len(n_parent)) {
  u <- rnorm(n_offspring[i], sd = sigma_parallel)
  v <- rnorm(n_offspring[i], sd = sigma_perp)

  xy_i <- data.frame(
    x = xy_parent$x[i] + u * cos(theta0) - v * sin(theta0),
    y = xy_parent$y[i] + u * sin(theta0) + v * cos(theta0)
  )
  xy <- rbind(xy, xy_i)
}

keep <- inside.owin(xy$x, xy$y, W) #only keep points in the window
X <- ppp(x = xy$x[keep], y = xy$y[keep], window = W)
```

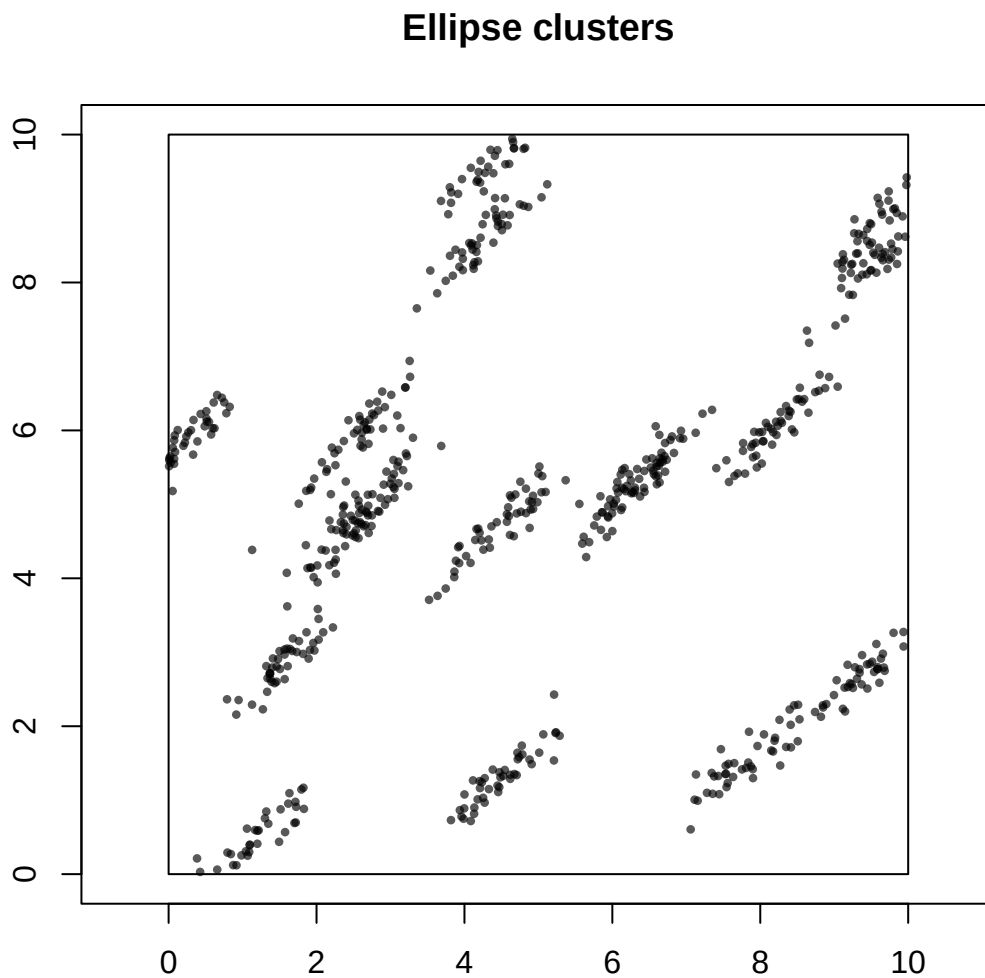
```
sum(n_offspring)
```

```
## [1] 716
```

```
npoints(X)
```

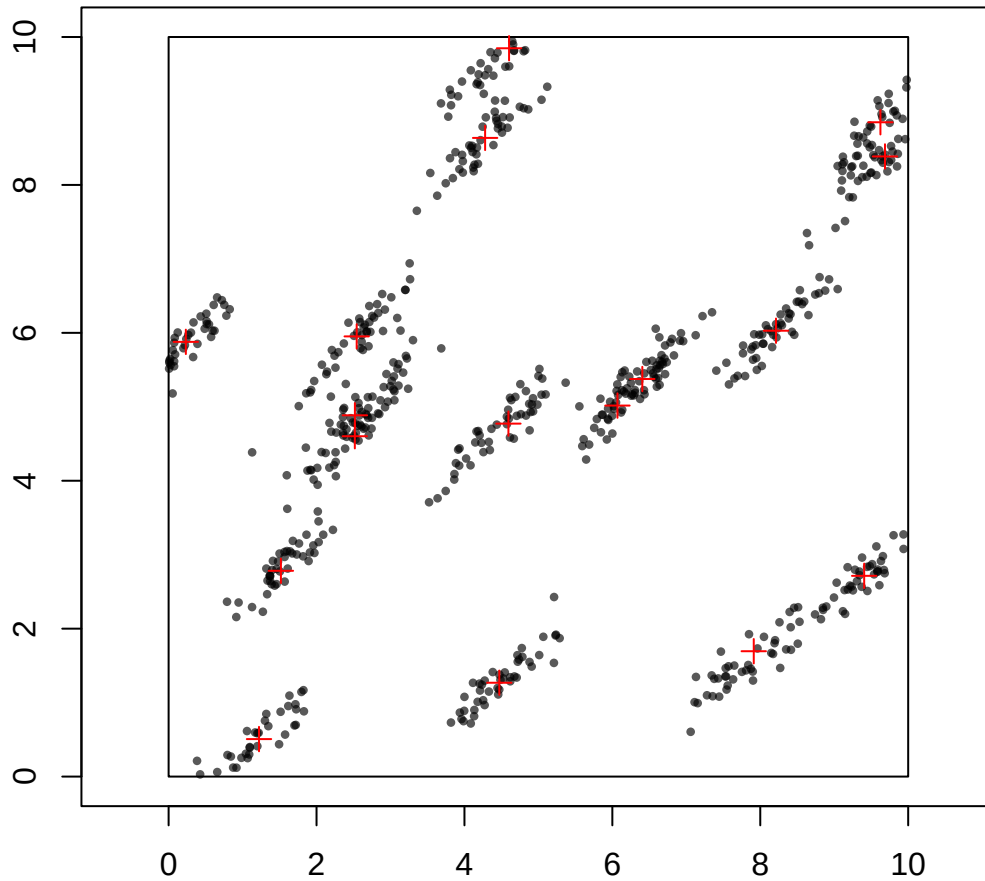
```
## [1] 671
```

```
plot(X, main = "Ellipse clusters", axes = TRUE, pch = 16, cex = 0.6)
```



```
plot(X, main = "Points and centres", axes = TRUE, pch = 16, cex = 0.6)  
plot(X_parent, add = TRUE, pch = 3, cols = "red", cex = 1.2)
```

Points and centres



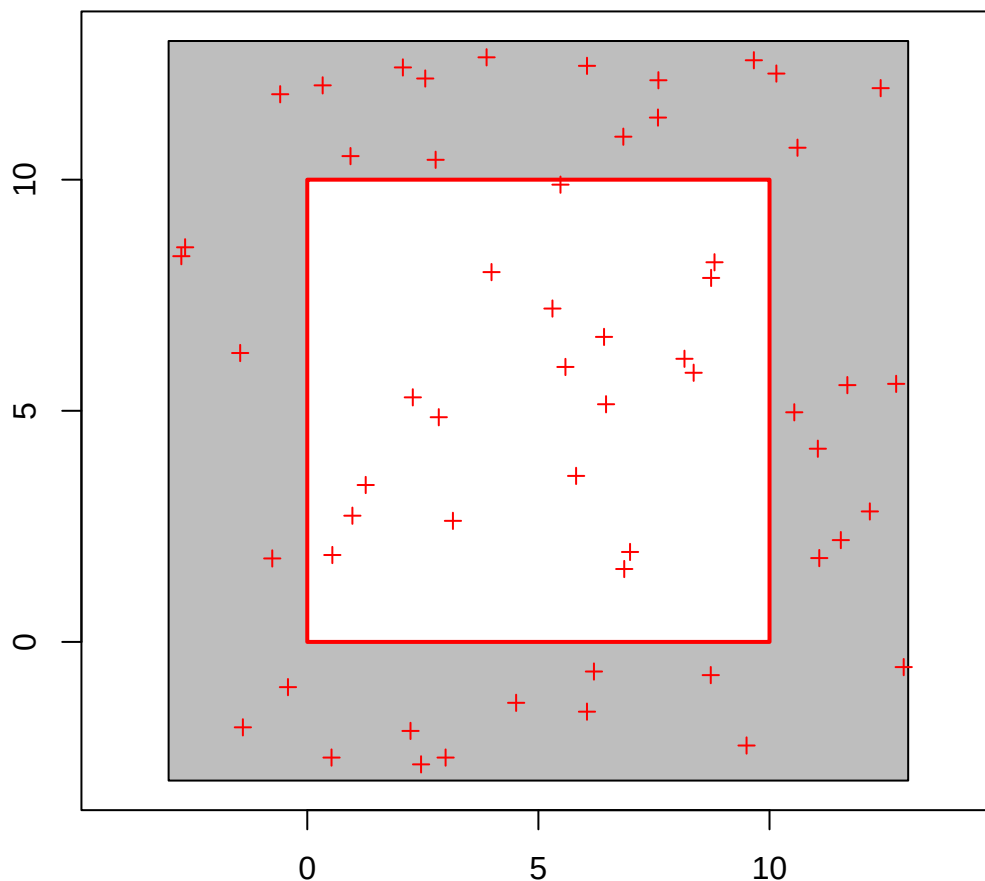
```
buffer <- 5 * max(sigma_parallel, sigma_perp)
W_sim <- owin(
  xrange = W$xrange + c(-buffer, buffer),
  yrange = W$yrange + c(-buffer, buffer)
)
```

```
X_parent <- rpoispp(lambda_centers, win = W_sim)
xy_parent <- data.frame(x = X_parent$x, y = X_parent$y)
n_parent <- npoints(X_parent)
n_parent
```

```
## [1] 57
```

```
plot(W_sim, main = "Observation window and buffer", axes = TRUE, col = "grey")
plot(W, add = TRUE, col = "white", border = "red", lwd = 2)
plot(X_parent, add = TRUE, pch = 3, cols = "red", cex = 0.8)
```

Observation window and buffer



```
sigma_gp <- 0.6
rho_gp <- 3

D_gp <- as.matrix(dist(xy_parent))
Sigma_gp <- sigma_gp^2 * exp(-D_gp^2 / (2 * rho_gp^2)) #Covariance

Sigma_gp <- sigma_gp^2 * exp(-D_gp^2 / (2 * rho_gp^2))
R_gp <- chol(Sigma_gp)

z_gp <- rnorm(n_parent)
Z_gp <- as.vector(t(R_gp) %*% z_gp)

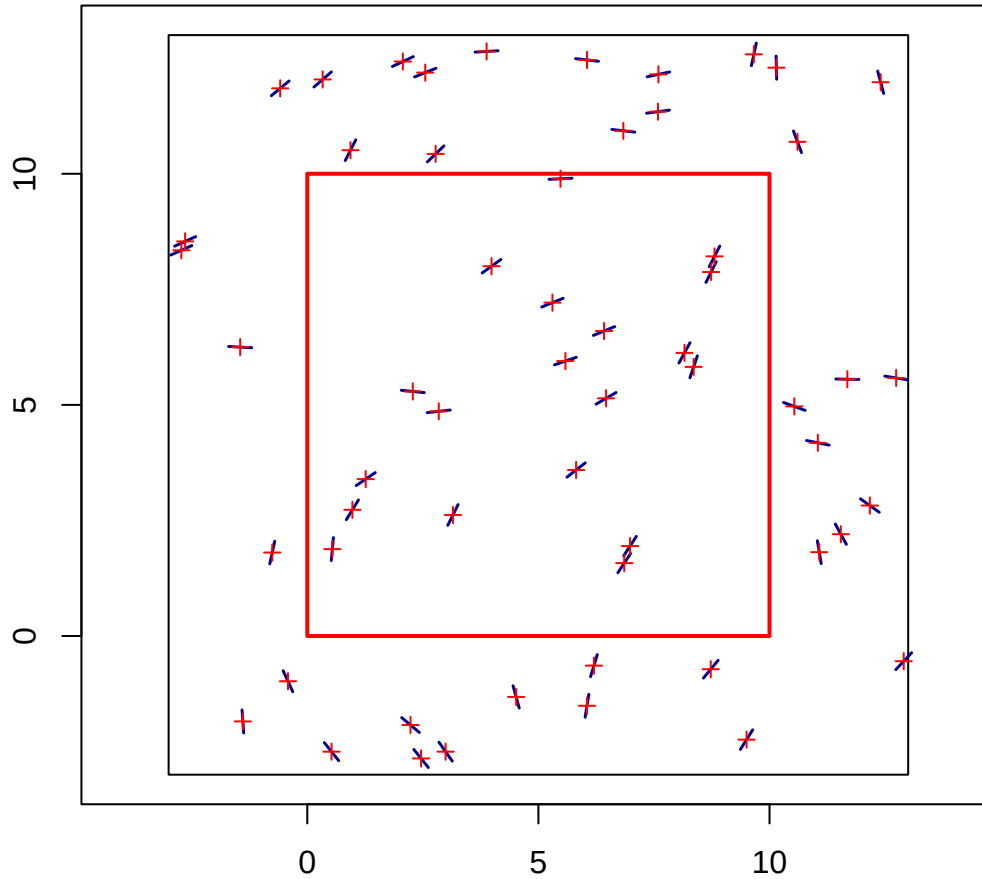
theta_gp <- (theta0 + Z_gp) %% pi

line_length <- 0.5
dx_parent <- line_length / 2 * cos(theta_gp)
dy_parent <- line_length / 2 * sin(theta_gp)

plot(W_sim, main = "GP directions at parents", axes = TRUE)
plot(W, add = TRUE, border = "red", lwd = 2)
segments(xy_parent$x - dx_parent, xy_parent$y - dy_parent,
         xy_parent$x + dx_parent, xy_parent$y + dy_parent,
         col = "darkblue", lwd = 1.5)
```

```
plot(X_parent, add = TRUE, pch = 3, cols = "red", cex = 0.8)
```

GP directions at parents



```
n_offspring <- rpois(n_parent, lambda = lambda_offspring)
n_offspring
```

```
## [1] 53 38 42 51 45 39 41 47 40 43 38 22 38 47 39 29 38 31 41 44 46 32 45 32 44
## [26] 42 43 39 38 46 38 48 39 30 56 44 43 42 37 39 44 33 37 32 34 42 37 49 39 43
## [51] 41 39 32 36 38 46 44
```

```
xy <- data.frame(x = numeric(), y = numeric())

for (i in seq_len(n_parent)) {
  u <- rnorm(n_offspring[i], sd = sigma_parallel)
  v <- rnorm(n_offspring[i], sd = sigma_perp)
  xy_i <- data.frame(
    x = xy_parent$x[i] + u * cos(theta_gp[i]) - v * sin(theta_gp[i]),
    y = xy_parent$y[i] + u * sin(theta_gp[i]) + v * cos(theta_gp[i])
  )
  xy <- rbind(xy, xy_i)
}

X <- ppp(x = xy$x, y = xy$y, window = W)
```

```
## Warning: 1502 points were rejected as lying outside the specified window
```

```
sum(n_offspring)
```

```
## [1] 2295
```

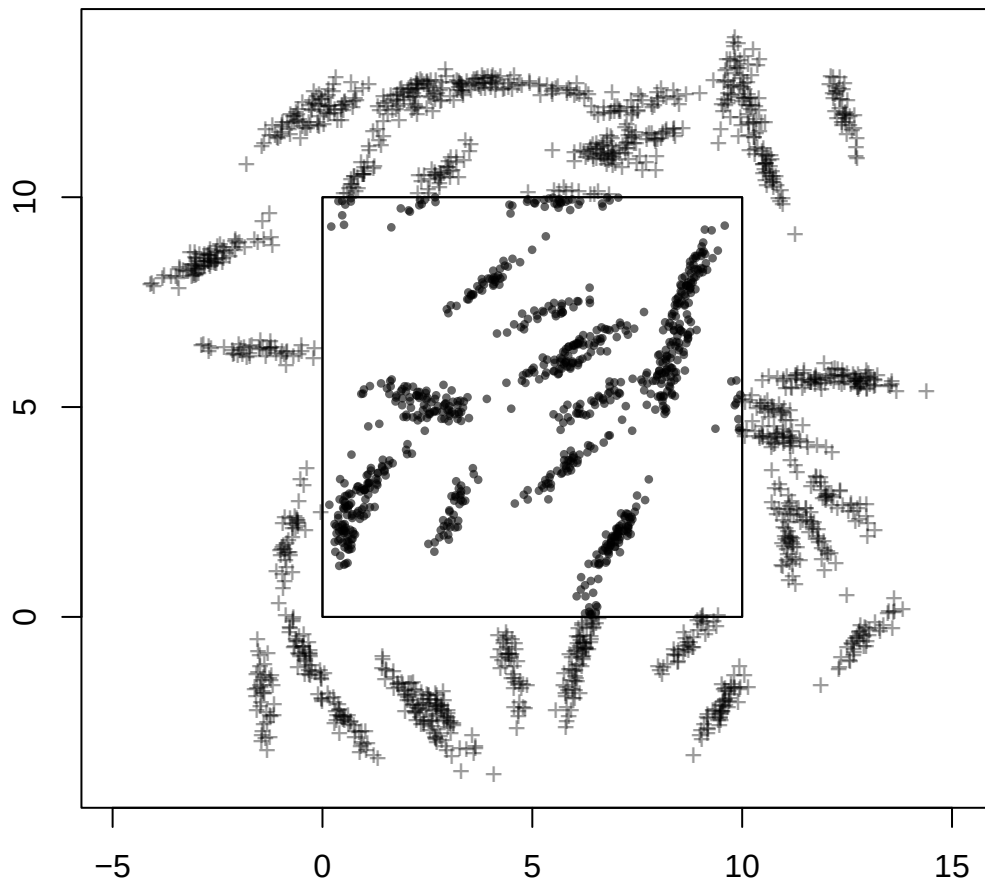
```
npoints(X)
```

```
## [1] 793
```

```
plot(X, main = "GP-directed", axes = TRUE, pch = 16, cex = 0.6)
```

```
## Warning in plot.ppp(X, main = "GP-directed", axes = TRUE, pch = 16, cex = 0.6):  
## 1502 illegal points also plotted
```

GP-directed



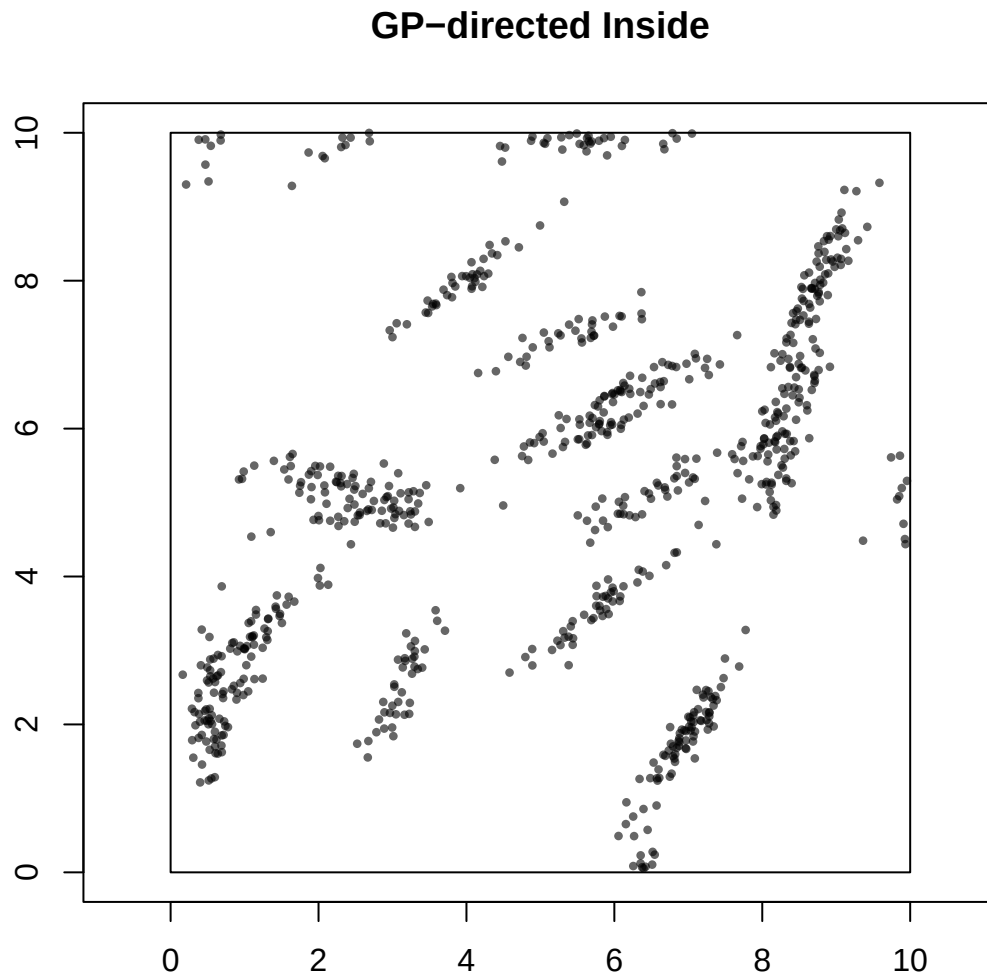
```
keep <- inside.owin(xy$x, xy$y, W)  
X <- ppp(x = xy$x[keep], y = xy$y[keep], window = W)  
sum(n_offspring)
```

```
## [1] 2295
```

```
npoints(X)
```

```
## [1] 793
```

```
plot(X, main = "GP-directed Inside", axes = TRUE, pch = 16, cex = 0.6)
```



```
plot(X, main = "GP-directed Inside", axes = TRUE, pch = 16, cex = 0.6)  
plot(X_parent[W], add = TRUE, pch = 3, cols = "red", cex = 0.8)
```

GP-directed Inside

